Y9

Tree Diagrams (Independent Events)

1). Each morning Bob and Bill catch the same bus. The probability that Bob catches the bus is 0.9 and for Bill it is 0.7. The probabilities are independent of each other.

a). Copy and **complete** the tree diagram.

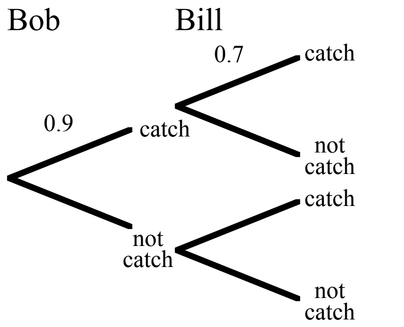
b). Calculate the probability that on a given day

i). they both catch the bus,

ii). Bob catches the bus, but not Bill,

iii). neither catch the bus,

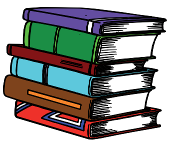
iv). at least one of them catches the bus.



2). There are 10 books on a shelf in a library. Seven are fiction and three are non-fiction. A member of the public takes a book at random, looks at it, and then replaces it on the shelf. Another member of the public then takes a book at random from the shelf.

a). Copy and **complete** the tree diagram.

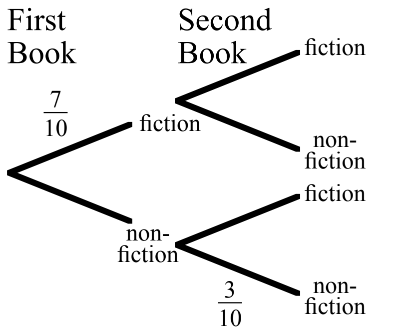
b). What is the probability that the two books taken are



i). both non-fiction,

ii). both fiction,

iii). one of each?



3). The Post Office have stated that 94% of all first class letters are delivered the next day.
Ron posts 2 letters on Friday.

a). Copy and **complete** the tree diagram.

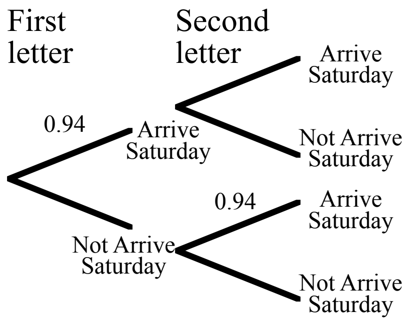
b). What is the probability that

i). both letters are delivered on Saturday,

ii). neither letter is delivered on Saturday,

iii). just one of the letters arrives on Saturday,

iv). at least one letter is delivered on Saturday?



4). In a box there are 6 red and 4 green counters, all the same size. One is drawn and then replaced. A second counter is then drawn.

a). Copy and **complete** the tree diagram.

b). What is the probability that

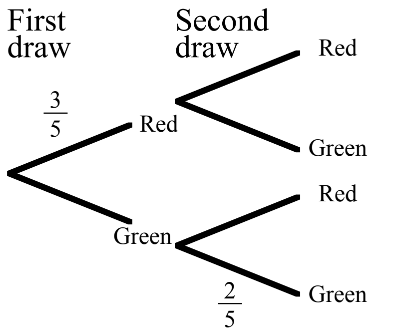
i). both counters are red,

ii). both counters are green,

iii). one is red and one green,

iv). at least one is green,

v). neither is green?



5). A fair coin is thrown three times.
Copy and **complete** the tree diagram, marking the probabilities and outcomes.
What is the probability that

a). all three coins are Heads,

b). exactly two coins land on Tails,

c). at least one coin lands on Tails,

d). no coin lands on Heads?